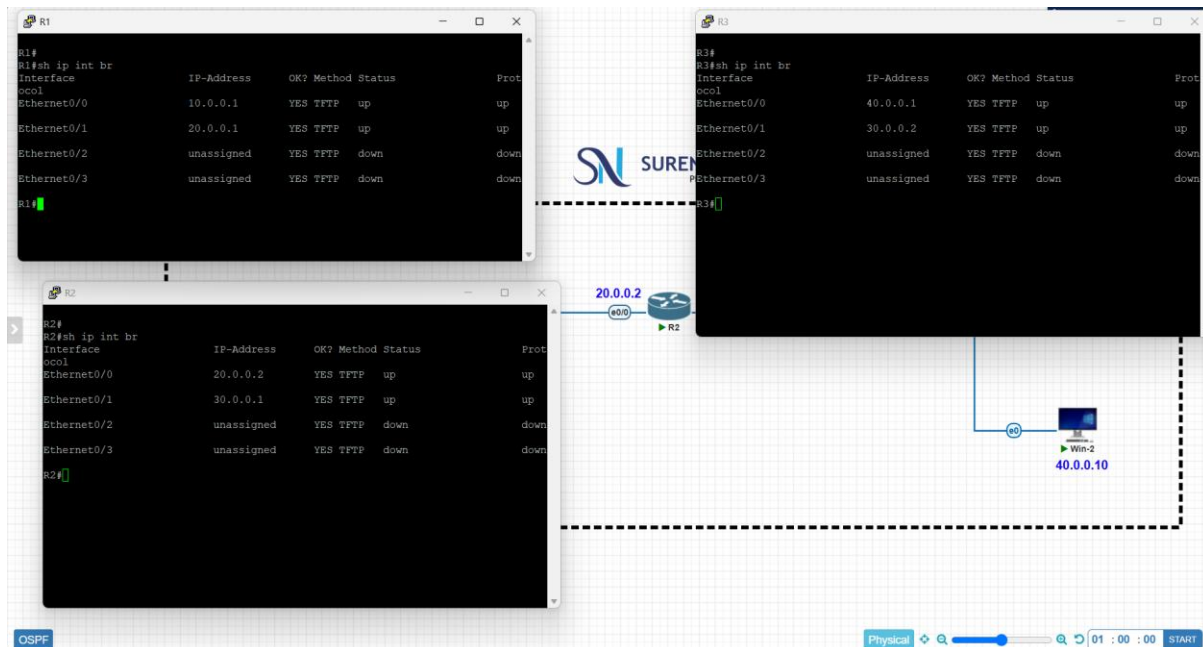


LAB 3 OPEN SHORTEST PATH FIRST PROTOCOL

GURUCHANDHRAN M

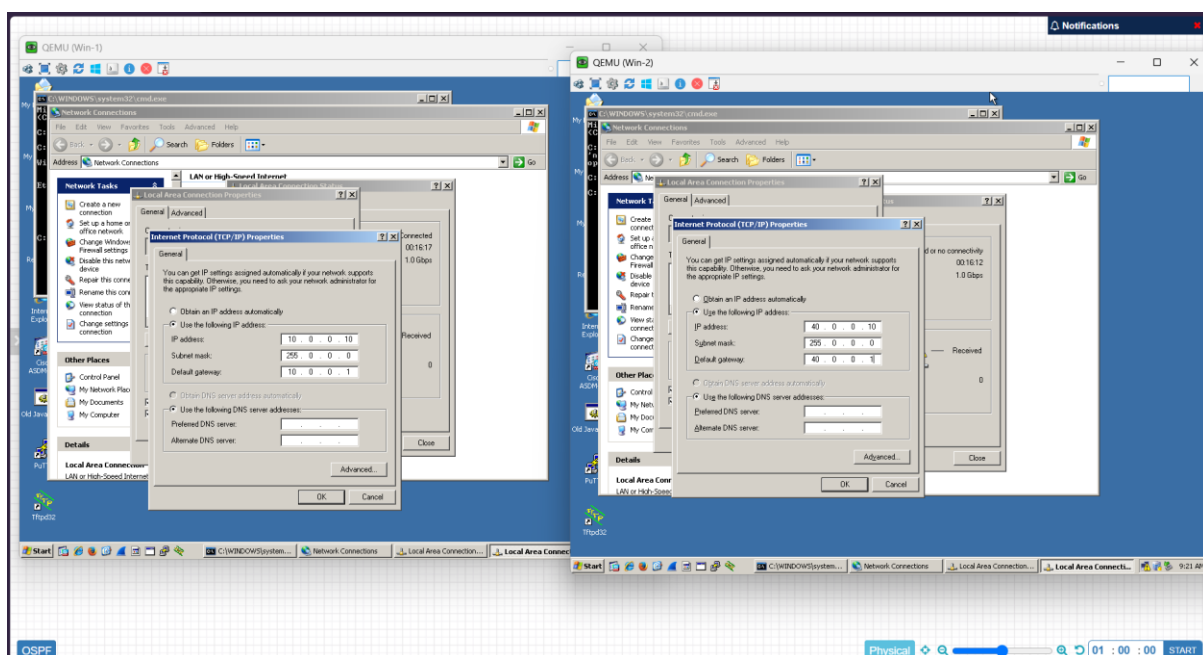
TASK 1: CHECKING ROUTER IP CONFIGURATIONS



TASK 2: PC IP ASSIGNMENT

PC1 IP: 10.0.0.10

PC2 IP: 40.0.0.10



TASK 3: ENABLING OSPF IN ROUTERS

R1 Configuration:

```

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#network-id 1.1.1.1
R1(config-router)#network 10.0.0.0 0.255.255.255 a 1
R1(config-router)#network 20.0.0.0 0.255.255.255 a 1
R1(config-router)#exit
R1(config)#
R1#

```

R2 Configuration:

```

R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#router-id 2.2.2.2
R2(config-router)#network 20.0.0.0 0.255.255.255 a
R2(config-router)#network 30.0.0.0 0.255.255.255 a 1
R2(config-router)#exit
R2(config)#
R2#

```

R3 Configuration:

```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#router-id 3.3.3.3
R3(config-router)#network 30.0.0.0 0.255.255.255 a 1
R3(config-router)#network 40.0.0.0 0.255.255.255 a 1
R3(config-router)#exit
R3(config)#
R3#

```

Network Diagram:

```

graph LR
    R1[10.0.0.10] --- R2[20.0.0.2]
    R2 --- R3[40.0.0.10]

```

TASK 4: VERIFYING OSPF NEIGHBORS

R1 Verification:

```

R1(config)#do sh ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
2.2.2.2 1 FULL/DR 00:00:35 20.0.0.2 Ethernet0/1
R1(config)#do sh ip ospf database
OSPF Router with ID (1.1.1.1) (Process ID 1)
Router Link States (Area 1)
Link ID ADV Router Age Seq# Checksum Link count
1.1.1.1 1.1.1.1 179 0x80000005 0x001788 2
2.2.2.2 2.2.2.2 125 0x80000005 0x003858 2
3.3.3.3 3.3.3.3 88 0x80000005 0x008606 2
Net Link States (Area 1)
Link ID ADV Router Age Seq# Checksum
20.0.0.1 1.1.1.1 179 0x80000001 0x00F81C
30.0.0.1 2.2.2.2 125 0x80000001 0x00AC52
R1(config)#do sh ip ospf int e0/0

```

R2 Verification:

```

R2(config)#do sh ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/DR 00:00:32 30.0.0.2 Ethernet0/1
1.1.1.1 1 FULL/DR 00:00:37 20.0.0.1 Ethernet0/0
R2(config)#do sh ip ospf database
OSPF Router with ID (2.2.2.2) (Process ID 1)
Router Link States (Area 1)
Link ID ADV Router Age Seq# Checksum Link count
1.1.1.1 1.1.1.1 252 0x80000005 0x001788 2
2.2.2.2 2.2.2.2 196 0x80000005 0x003858 2
3.3.3.3 3.3.3.3 159 0x80000005 0x008606 2
Net Link States (Area 1)
Link ID ADV Router Age Seq# Checksum
20.0.0.1 1.1.1.1 252 0x80000001 0x00F81C
30.0.0.1 2.2.2.2 196 0x80000001 0x00AC52
R2(config)#

```

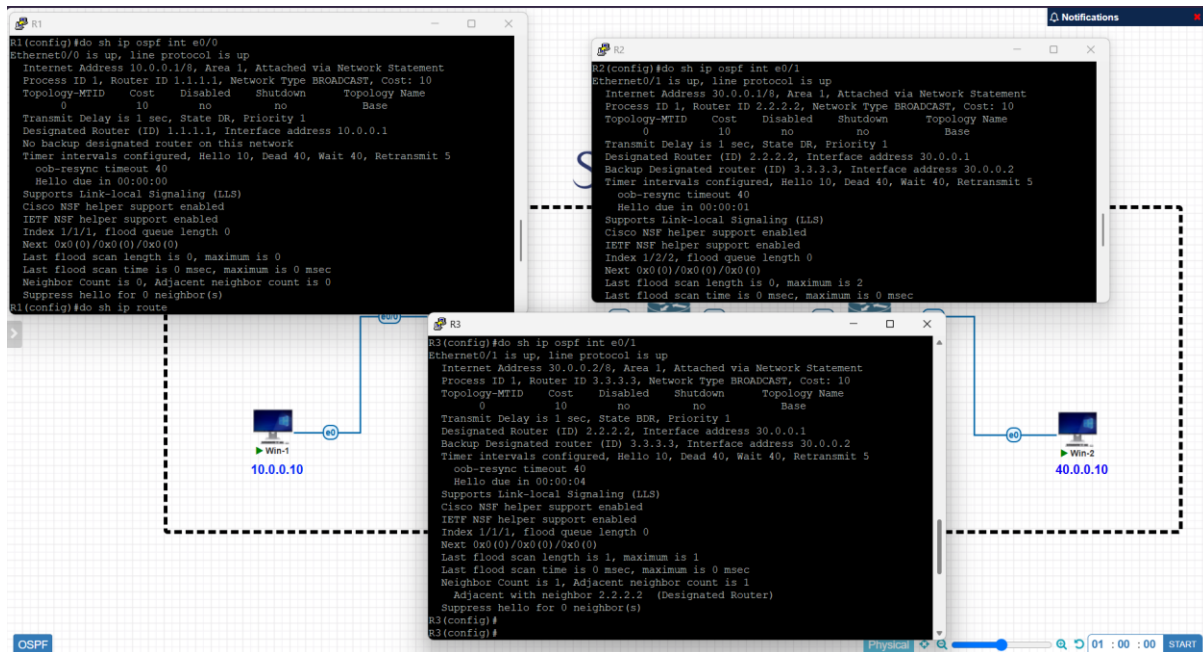
R3 Verification:

```

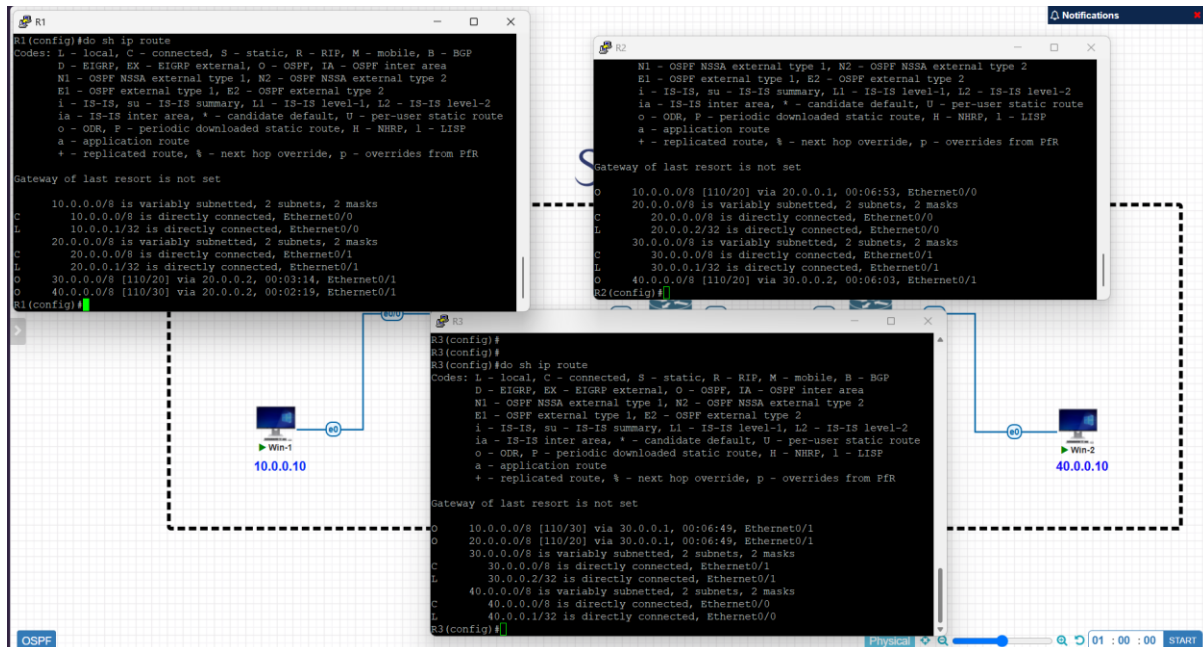
R3(config)#do sh ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
2.2.2.2 1 FULL/DR 00:00:30 30.0.0.1 Ethernet0/1
R3(config)#do sh ip ospf database
OSPF Router with ID (3.3.3.3) (Process ID 1)
Router Link States (Area 1)
Link ID ADV Router Age Seq# Checksum Link count
1.1.1.1 1.1.1.1 275 0x80000005 0x001788 2
2.2.2.2 2.2.2.2 220 0x80000005 0x003858 2
3.3.3.3 3.3.3.3 181 0x80000005 0x008606 2
Net Link States (Area 1)
Link ID ADV Router Age Seq# Checksum
20.0.0.1 1.1.1.1 275 0x80000001 0x00F81C
30.0.0.1 2.2.2.2 220 0x80000001 0x00AC52
R3(config)#

```

TASK 5: VERIFYING OSPF IN THE INTERFACES



TASK 6: VERIFYING ROUTES IN THE ROUTERS



TASK 7: VERIFYING CONNECTION ESTABLISHMENT USING ping

